

Medical Statistics – Answers lab 7

Part 1: Risk of in-hospital death in patients with acute myocardial infarction

Exploratory data analysis

	alive	dead
male	282	18
female	179	21

	alive	dead
male	0.940	0.060
female	0.895	0.105

Question 1

Based on the group proportions, do you expect gender to have an effect on the risk of in-hospital death?

Answer question 1

The proportion of in-hospital deaths is higher among females (0.105) than among males (0.06). This suggests that gender might be associated with the risk of in-hospital death.

Simple logistic regression

Question 2

Perform the chi-square test of homogeneity (see instructions in lab 4 if needed). What conclusion can be drawn from the test?

Pearson's Chi-squared test

```
data: contingencyTable  
X-squared = 3.3789, df = 1, p-value = 0.06603
```

Answer question 2

The p-value of the chi-square test (0.066) is larger than 0.05, indicating that there is no significant association between gender and in-hospital death.

Call:

```
glm(formula = dstat_numeric ~ gender, family = binomial, data = whas500)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-2.7515	0.2431	-11.318	<2e-16 ***
genderfemale	0.6087	0.3351	1.816	0.0693 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 273.86 on 499 degrees of freedom
Residual deviance: 270.55 on 498 degrees of freedom
AIC: 274.55

Number of Fisher Scoring iterations: 5

Question 3

What is the odds ratio for in-hospital death for females compared to males? How should this odds ratio be interpreted in the context of the study?

Answer question 3

The odds ratio for in-hospital death for females compared to males is equal to $\exp(0.6087) = 1.83$. This means that the odds of in-hospital death are 1.83 times higher in females than in males.

Question 4

Based on the estimated regression coefficients (ignoring p-values), what are the predicted proportions of in-hospital deaths for male and female patients? Compare the predicted proportions to the observed proportions from the previously constructed contingency table. Do they match?

Answer question 4

The linear predictor models the risk of in-hospital death on the log-odds scale. To obtain the predicted probabilities, we need to transform the log-odds back to the probability scale using the logistic function.

The predicted probabilities of in-hospital death on the log-odds scale are:

- Males: -2.7515 (intercept)
- Females: $-2.7515 + 0.6087$ (intercept + coefficient) = -2.1428

The predicted probabilities on the probability scale are:

- Male: $1/(1 + \exp(-(-2.7515))) = 0.060$
- Female: $1/(1 + \exp(-(-2.1428))) = 0.105$

The predicted proportions of in-hospital deaths match the observed proportions from the contingency table.

Question 5

What conclusion can be drawn from the logistic regression analysis regarding the association between gender and in-hospital death? Is this in line with the conclusion drawn from the chi-square test?

Answer question 5

The p-value of the Wald test for the regression coefficient of the dummy variable `genderfemale` is 0.069, which is larger than 0.05. This indicates that there is no significant association between gender and in-hospital death. This conclusion is consistent with the results of the chi-square test.

Multiple logistic regression

Call:

```
glm(formula = dstat_numeric ~ gender + age, family = binomial,  
     data = whas500)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-6.89600	1.19131	-5.789	7.1e-09 ***
genderfemale	0.26776	0.34618	0.773	0.439246
age	0.05770	0.01524	3.785	0.000154 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 273.86 on 499 degrees of freedom
Residual deviance: 253.45 on 497 degrees of freedom
AIC: 259.45

Number of Fisher Scoring iterations: 6

Question 6

How does adjusting for age affect the estimated odds ratio for in-hospital death for females compared to males?

Answer question 6

After adjusting for age, the odds ratio for in-hospital death for females compared to males decreases from 1.83 to $\exp(0.26776) = 1.31$. This indicates that the effect of gender on in-hospital death is confounded by age.

Question 7

Calculate the odds ratio for in-hospital death corresponding to a 10-year increase in age and interpret its meaning.

Answer question 7

The increase in the risk of in-hospital death on the log-odds scale is equal to $0.0577 * 10 = 0.577$. This means that for every 10-year increase in age, the odds of in-hospital death increase by a factor of $\exp(0.577) = 1.78$.

Likelihood ratio tests

Analysis of Deviance Table

```
Model 1: dstat_numeric ~ age
Model 2: dstat_numeric ~ gender + age
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1      498      254.05
2      497      253.45  1   0.59934   0.4388
```

Question 8

How does the p-value from the likelihood ratio test compare to the one from the Wald test?

Answer question 8

The p-value of the likelihood ratio test (0.4388) is approximately equal to the p-value of the Wald test (0.4392). This indicates that the results of the two tests are consistent, which is expected because, for testing the significance of a single coefficient, the Wald test and the likelihood ratio test often yield similar results. However, they are not strictly equivalent, as the likelihood ratio test is based on comparing model fit, while the Wald test assesses the coefficient's deviation from zero using its standard error.

Evaluating model fit

R results

Groups for Hosmer-Lemeshow C statistic:

```

cutfit
[0.00568,0.0178] (0.0178,0.0279] (0.0279,0.0369] (0.0369,0.0514]
                    55                    52                    47                    52
(0.0514,0.065] (0.065,0.0864] (0.0864,0.103] (0.103,0.126]
                    44                    53                    55                    43
(0.126,0.154] (0.154,0.348]
                    50                    49

```

Groups for Hosmer-Lemeshow H statistic:

```

cutfit1
[0.00534,0.0399] (0.0399,0.0742] (0.0742,0.108] (0.108,0.143]
                    164                    110                    87                    61
(0.143,0.177] (0.177,0.211] (0.211,0.245] (0.245,0.28]
                    54                    19                    2                    2
(0.28,0.314] (0.314,0.348]
                    0                    1

```

\$C

Hosmer-Lemeshow C statistic

```

data: fitted(model.sex.age) and model.sex.age$y
X-squared = 11.465, df = 8, p-value = 0.1767

```

\$H

Hosmer-Lemeshow H statistic

```

data: fitted(model.sex.age) and model.sex.age$y
X-squared = 6.2434, df = 8, p-value = 0.62

```

SPSS results

Hosmer and Lemeshow Test			
Step	Chi-square	df	Sig.
1	12,771	8	,120

Contingency Table for Hosmer and Lemeshow Test						
		discharge status from hospital = alive		discharge status from hospital = dead		Total
		Observed	Expected	Observed	Expected	
Step 1	1	49	48,372	0	,628	49
	2	48	48,906	2	1,094	50
	3	47	48,409	3	1,591	50
	4	49	46,893	0	2,107	49
	5	51	48,992	1	3,008	52
	6	42	44,420	6	3,580	48
	7	44	46,250	7	4,750	51
	8	49	46,047	3	5,953	52
	9	45	42,849	5	7,151	50
	10	37	39,861	12	9,139	49

Figure 1: Screenshot of the Hosmer-Lemeshow test results in SPSS

Question 9

Based on the results of the Hosmer-Lemeshow goodness-of-fit test, does our model provide a satisfactory fit to the data?

Answer question 9

The p-value for the Hosmer-Lemeshow test is 0.177 in R (C statistic) and 0.120 in SPSS, indicating that the model provides a satisfactory fit to the data: a non-significant p-value suggests that the model fits the data well, meaning that the observed and expected frequencies do not differ significantly.

Note: the discrepancy between the p-values from R and SPSS seems to be due to small differences in how the groups are created within the implementation of the test.

Part 2: unguided exercises

Exercise 1

Multiple logistic regression was used to construct a prognostic index to predict coronary artery disease from data on 348 patients with valvular heart disease who had undergone routine coronary arteriography before valve replacement (Ramsdale et al. 1982). The estimated equation was:

$$\text{logit}(p) = \ln(p/(1-p)) = b_0 + 1.167 \times x_1 + 0.0106 \times x_2 + \text{other terms}$$

where x_1 stands for the family history of ischaemic disease (0=no, 1=yes) and x_2 is the estimated total number of cigarettes ever smoked in terms of thousand cigarettes, calculated as the average number smoked annually times the number of years smoking.

- (a) What is the estimated odds ratio for having coronary artery disease for subjects with a positive family history relative to subjects with a negative family history?
- (b) What total number of cigarettes ever smoked carries the same risk as a positive family history? Convert the result into years of smoking 20 cigarettes per day.
- (c) What is the odds ratio for coronary artery disease for someone with a positive family history who had smoked 20 cigarettes a day for 30 years compared to a non smoker with no family history?

Answers:

- (a) $\exp(1.167) = 3.212$.
- (b) The $\log(\text{OR})$ of positive history, 1.167, is to be set equal to $0.0106 \times x_2$, where 0.0106 is the log odds ratio of smoking 1000 cigarettes. Thus $x_2 = 1.167/0.0106 = 110.094$ thousands of cigarettes. Dividing this result by $(365 \times 20)/1000 = 7.3$, i.e., the total number cigarettes (per thousand) smoked in 1 year if smoking 20 cigarettes per day, we find $110.094/7.3 = 15.1$ or just above 15 years. Thus the odds ratio of positive history is equivalent to that of daily smoking of 20 cigarettes for about 15 years.
- (c) The total number of cigarettes smoked (per thousand) is $(20 \times 365 \times 30)/1000 = 219$, so the odds ratio is $\exp(1.167 + 219 \times 0.0106) = 32.7$

Exercise 2

Data from 37 patients receiving a non-depleted allogenic bone marrow transplant were examined to see which variables were associated with the occurrence of acute graft-versus-host disease (GvHD: 0=no, 1=yes) (Bagot et al., 1988). Possible predictors are TYPE (type of leukemia: 1=AML, acute myeloid leukaemia; 2=ALL, acute lymphocytic leukaemia; 3=CML, chronic

myeloid leukemia), PREG (donor pregnancy: 0= no, 1=yes), and LOGIND (the logarithm of an index of mixed epidermal cell-lymphocyte reactions). The data are in the file `GvHD.sav` available on Brightspace.

- (a) Perform a likelihood ratio test to determine whether there is a significant association between the type of leukemia and the occurrence of GvHD after adjusting for donor pregnancy and the logarithm of an index of mixed epidermal cell-lymphocyte reactions.
- (b) In the adjusted model, What is the estimated odds ratio for the occurrence of GvHD for patients with ALL compared to those with ALM?
- (c) Use the Hosmer-Lemeshow goodness-of-fit test to evaluate the fit of the model. Based on the results, does the model provide a satisfactory fit to the data?

Answers:

```
# Create a 0/1 numeric version of the outcome variable
GvHD$gvhd.numeric <- ifelse(GvHD$gvhd == "no", 0, 1)

# Fit the full model
model.GvHD <- glm(gvhd.numeric ~ type + preg + logind, family = binomial, data = GvHD)
summary(model.GvHD)
```

Call:

```
glm(formula = gvhd.numeric ~ type + preg + logind, family = binomial,
     data = GvHD)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-2.4496	1.2113	-2.022	0.0431 *
typeALL	-0.1480	1.1722	-0.126	0.8996
typeCML	2.1781	1.2420	1.754	0.0795 .
pregyes	2.4982	1.1026	2.266	0.0235 *
logind	1.4577	0.7547	1.932	0.0534 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 51.049 on 36 degrees of freedom
 Residual deviance: 28.832 on 32 degrees of freedom
 AIC: 38.832

Number of Fisher Scoring iterations: 5

```
# Fit the reduced model
model.GvHD.reduced <- glm(gvhd.numeric ~ preg + logind, family = binomial, data = GvHD)

# Perform likelihood ratio test
anova(model.GvHD.reduced, model.GvHD, test = "LRT")
```

Analysis of Deviance Table

```
Model 1: gvhd.numeric ~ preg + logind
Model 2: gvhd.numeric ~ type + preg + logind
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1      34      33.654
2      32      28.832  2    4.822  0.08972 .
---
```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

- The p-value of the likelihood ratio test is 0.090, indicating that the type of leukemia is not significantly associated with the occurrence of GvHD after adjusting for donor pregnancy and the logarithm of an index of mixed epidermal cell-lymphocyte reactions.
- By default, R and SPSS both use the first level of the factor (ALM) as the reference category. Therefore, the table with estimated regression coefficients is the same in both outputs. The estimated odds ratio for the occurrence of GvHD for patients with AML compared to those with ALL is $\exp(-0.148) = 0.862$.
- The Hosmer-Lemeshow test results are as follows:

R

Groups for Hosmer-Lemeshow C statistic:

```
cutfit
[0.0109,0.0254] (0.0254,0.0552] (0.0552,0.142] (0.142,0.237] (0.237,0.529]
      4          4          3          4          4
(0.529,0.582] (0.582,0.756] (0.756,0.879] (0.879,0.891] (0.891,0.988]
      3          4          3          4          4
```

Groups for Hosmer-Lemeshow H statistic:

```
cutfit1
[0.00994,0.109] (0.109,0.206] (0.206,0.304] (0.304,0.402] (0.402,0.5]
      11          2          3          1          1
(0.5,0.597] (0.597,0.695] (0.695,0.793] (0.793,0.89] (0.89,0.989]
      5          0          4          6          4
```

\$C

Hosmer-Lemeshow C statistic

```
data: fitted(model.GvHD) and GvHD$gvhd.numeric  
X-squared = 12.352, df = 8, p-value = 0.1362
```

\$H

Hosmer-Lemeshow H statistic

```
data: fitted(model.GvHD) and GvHD$gvhd.numeric  
X-squared = 14.907, df = 8, p-value = 0.06097
```

SPSS

Hosmer and Lemeshow Test						
Step	Chi-square	df	Sig.			
1	12,465	7	,086			

Contingency Table for Hosmer and Lemeshow Test						
		GvHD = no		GvHD = yes		Total
		Observed	Expected	Observed	Expected	
Step 1	1	4	3,929	0	,071	4
	2	4	3,843	0	,157	4
	3	4	3,628	0	,372	4
	4	4	3,142	0	,858	4
	5	0	2,151	4	1,849	4
	6	1	1,554	3	2,446	4
	7	1	,945	3	3,055	4
	8	2	,484	2	3,516	4
	9	0	,323	5	4,677	5

Figure 2: Screenshot of the Hosmer-Lemeshow test results in SPSS

While R and SPSS produce different p-values, neither is significant. Therefore, the model provides a satisfactory fit to the data.